

Application circuits

A commonly used application circuit employing PIN photodiode in photovoltaic mode is shown in Fig. 6. The output voltage is given by $I_{det} \times R$, where I_{det} is the current through the photodiode.

Fig. 7 shows four possible circuits of PIN photodiodes in photoconductive mode. Fig. 7(a) shows the basic circuit with no amplification. In Fig. 7(b) the operational amplifier is used as a voltage amplifier, whereas in Figs 7(c) and (d) the operational amplifier

is used in transimpedance mode. For the circuit in Fig. 7(b), output voltage and effective resistance across the photodiode are $(I_{det} \times R)$ and R , respectively. I_{det} is the current flowing through the photodiode. Output voltage and effective resistance across the photodiode in Figs 7(c) and (d) are $(I_{det} \times R_f)$ and R_f/A , respectively, where I_{det} is photodiode current and A is open-loop gain of the operational amplifier. Circuits in Figs 7(c) and (d) offer better linearity.

Avalanche photodiodes are also connected in a similar manner as discussed above, except that a much higher reverse bias voltage is required for their operation. Also, their power consumption during operation is much higher than PIN photodiodes and is given by the product of input signal, sensitivity and reverse bias voltage. Hence a protective resistor is added to the bias circuit (Fig. 8) or a current-limiting circuit is used.

Solar cells

Solar cells operate very similar to photodiodes in photovoltaic mode. These operate on the principle of photovoltaic effect. As mentioned above, due to photovoltaic effect, an open-circuit voltage develops

across a p-n junction when it is exposed to light, which is solar radiation in the case of a solar cell. This open-circuit voltage leads to the flow of electric current through a load resistance connected across it (Fig. 9).

As shown in Fig. 9, the impinging photon energy causes generation of electron-hole pairs. These electron-hole pairs either recombine and vanish, or start drifting in opposite directions with electrons moving towards n-region and holes moving towards p-region. The accumulation of positive and negative charge carriers constitutes the open-circuit voltage. This voltage can cause a current through the external load or, when the junction is shorted, a short-circuit current whose magnitude is proportional to the input light intensity.

The voltage output and the current delivery capability of an individual solar cell are very small for use as an electrical power input to any system. As an example, a typical solar cell would produce 500mV output with load current capability of about 150mA. A series-parallel arrangement of solar cells gives the desired output voltage with required power delivery capability. Series combination enhances the output voltage, while parallel combination enhances the current rating. Fig. 10 shows some representative solar cell panels.

Solar cell efficiency is the ratio of maximum electrical solar cell power to the radiant light power on the solar cell area. The efficiency figure for some crystalline solar cells is in excess of 20 per cent. Silicon is most commonly used semiconductor material for solar cells. Both crystalline and amorphous forms of silicon are used for the purpose. Another promising material for solar cells is gallium-arsenide (GaAs). Gallium-arsenide solar cells, when perfected, will be light-weight and more efficient.

Photoemissive sensors

Photoemissive sensors rely on external photo effect, wherein photo-generated electrons travel beyond photo-

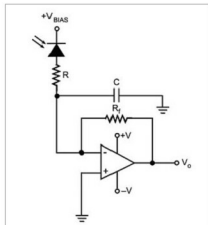


Fig. 8: Application circuit using avalanche photodiode

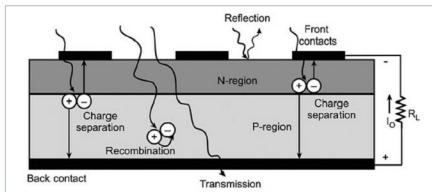


Fig. 9: Operating principle of a solar cell



Fig. 10: Different package styles of solar cells